

# Week 3 - Circom

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## What is Circom?

Circom is a domain-specific programming language for building zero-knowledge circuits. A circuit in this context is like a recipe that describes how to check that some computation was done correctly, without showing the private data involved. Circom helps developers write these circuits in a structured way, so they don't have to work directly with low-level cryptography.

## How Circom Works

When you write a program in Circom, you describe inputs, outputs, and the relationships between them. These relationships are turned into mathematical constraints. Later, a proof system (like SNARKs) can use those constraints to generate a proof that the computation was correct. The magic is that the verifier can check the proof without knowing the secret inputs. For example, you could prove that you know a password that unlocks an account without ever revealing the password itself.

## Circuits and Components

In Circom, a circuit is built from smaller parts called components. Each component defines a piece of logic, such as addition or hashing, and you can connect components together to form bigger circuits. This is similar to building with Lego blocks: small pieces combine into larger structures. Circom provides a standard library of useful components, and developers can also create their own.

## Workflow with Circom

The basic workflow has a few steps. First, you write your circuit in a `.circom` file. Then you compile it, which produces files that describe the constraints. Next, you set up the proving system, which may involve creating keys. Finally, you can generate a proof from a witness (the secret input) and verify that

proof. Tools like **snarkjs** are commonly used alongside Circom to handle proof generation and verification.

## Why Learn Circom?

Circom is important because it allows developers to design real-world applications that use zero-knowledge proofs. With Circom, you can build private voting systems, anonymous authentication methods, or blockchain applications that verify transactions without exposing sensitive information. It lowers the barrier to using advanced cryptography by giving developers a simple language and toolkit.

## Key Takeaways

Circom is a language designed to make building zero-knowledge circuits easier. It turns logic into mathematical constraints that proof systems can use. Circuits are made from reusable components, and the workflow goes from writing code to generating and verifying proofs. Learning Circom provides the foundation for building privacy-preserving applications in blockchain and beyond.